

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 6

Review: Lectures 12–13

- **Smooth constrained problem and Lagrangian.** For an open domain $\mathcal{D} \subseteq \mathbf{R}^n$,

$$\mathcal{S} = \{\mathbf{x} \in \mathcal{D} : g_i(\mathbf{x}) = 0, h_j(\mathbf{x}) \leq 0\}, \quad J(\mathbf{x}) = \{j : h_j(\mathbf{x}) = 0\}.$$

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x}).$$

The equality multipliers λ_i 's are unrestricted; the inequality multipliers ν_j 's are nonnegative.

- **KKT first-order conditions and KKT points.** A feasible point $\mathbf{x}^*$ is a *KKT point* if multipliers exist with

$$\begin{aligned} \nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) &= \mathbf{0} && \text{(stationarity),} \\ g_i(\mathbf{x}^*) &= 0, \quad h_j(\mathbf{x}^*) \leq 0 && \text{(primal feasibility),} \\ \nu_j^* &\geq 0 && \text{(dual feasibility),} \\ \nu_j^* h_j(\mathbf{x}^*) &= 0 && \text{(complementary slackness).} \end{aligned}$$

Only active inequalities can contribute to stationarity. In the force interpretation, the forces are $-\nabla f$, $-\lambda_i \nabla g_i$, and $-\nu_j \nabla h_j$.

- **LICQ and KKT first-order necessary conditions (FONC).** LICQ holds at $\mathbf{x} \in \mathcal{S}$ if the family

$$\{\nabla g_i(\mathbf{x}) : i = 1, \dots, m\} \cup \{\nabla h_j(\mathbf{x}) : j \in J(\mathbf{x})\} \quad \text{is linearly independent.}$$

For continuously differentiable data,

$$\mathbf{x}^* \text{ local minimizer} + \text{LICQ} \implies \mathbf{x}^* \text{ is a KKT point with unique multipliers.}$$

Warnings. A KKT point need not be a local minimizer; LICQ is sufficient, not necessary, for KKT multipliers to exist; an inactive inequality has $\nu_j^* = 0$, but an active inequality can also have $\nu_j^* = 0$ (contact without load).

Review: Lectures 12–13 (continued)

- **Tangent and linearized cones.**

$$T_{\mathcal{S}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{l} \exists t^{(k)} \downarrow 0, \mathbf{x}^{(k)} \in \mathcal{S}, \mathbf{x}^{(k)} \rightarrow \mathbf{x}^*, \\ (\mathbf{x}^{(k)} - \mathbf{x}^*)/t^{(k)} \rightarrow \mathbf{d} \end{array} \right\}.$$

$$T_{\text{lin}}(\mathbf{x}^*) = \{ \mathbf{d} : \nabla g_i(\mathbf{x}^*)^T \mathbf{d} = 0; \nabla h_j(\mathbf{x}^*)^T \mathbf{d} \leq 0 \ (j \in J(\mathbf{x}^*)) \}.$$

Always $T_{\mathcal{S}} \subseteq T_{\text{lin}}$; LICQ gives equality. At a local minimizer, $\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0$ for all $\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*)$.

- **Critical cone: first-order-neutral directions.**

$$C_{\text{crit}}(\mathbf{x}^*) = \{ \mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nabla f(\mathbf{x}^*)^T \mathbf{d} = 0 \}.$$

At a KKT triple, define $J_+ = \{j \in J(\mathbf{x}^*) : \nu_j^* > 0\}$ and $J_0 = \{j \in J(\mathbf{x}^*) : \nu_j^* = 0\}$. Then

$$C_{\text{crit}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x}^*)^T \mathbf{d} = 0, & i = 1, \dots, m, \\ \nabla h_j(\mathbf{x}^*)^T \mathbf{d} = 0, & j \in J_+, \\ \nabla h_j(\mathbf{x}^*)^T \mathbf{d} \leq 0, & j \in J_0 \end{array} \right\}.$$

Loaded walls impose equality; unloaded active walls retain inequality.

- **KKT second-order conditions.** For twice continuously differentiable data, set

$$\mathbf{H}_L = \nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*).$$

SONC: local minimizer + LICQ $\implies \mathbf{d}^T \mathbf{H}_L \mathbf{d} \geq 0$ for every $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$.

SOSC: KKT first-order conditions + $\mathbf{d}^T \mathbf{H}_L \mathbf{d} > 0$ for every nonzero $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ $\implies$ strict local minimum (indeed, quadratic growth).

Warnings. Use $\mathbf{H}_L$, not merely $\mathbf{H}_f$. SONC is not sufficient; SOSC is not necessary. If $C_{\text{crit}} = \{\mathbf{0}\}$, SOSC is vacuously satisfied.

- **Reduced Hessian and logical limits.** If $C_{\text{crit}} = \text{Range}(Z)$ is a subspace and Z has independent columns, SONC/SOSC reduce to $Z^T \mathbf{H}_L Z \succeq 0$ / $Z^T \mathbf{H}_L Z \succ 0$. This applies to equality-only problems and/or under strict complementarity ($J_0 = \emptyset$). A general cone is not a subspace.

1. Consider

$$\begin{aligned} \min \quad & f(\mathbf{x}) = -\frac{(x_1 - 1)^2}{16} - 4x_2^2 \\ \text{s.t.} \quad & h_1(\mathbf{x}) = -x_1 - 8x_2 - 3 \leq 0, \quad h_2(\mathbf{x}) = x_1 + 8x_2 - 5 \leq 0, \\ & h_3(\mathbf{x}) = -x_1 + 8x_2 - 3 \leq 0, \quad h_4(\mathbf{x}) = x_1 - 8x_2 - 5 \leq 0. \end{aligned}$$

It is given that all feasible points satisfy LICQ and that

$$\mathbf{a} = (-3, 0)^T, \quad \mathbf{b} = (5, 0)^T, \quad \mathbf{c} = (1, 1/2)^T$$

are global minimizers.

- (a) Find the active sets and verify the KKT FONC at these three points.
- (b) Find J_+ , J_0 , and the critical cone at each point. Explain why SOSC holds even though $\mathbf{H}_f \prec 0$ on $\mathbf{R}^2$.
- (c) Keep the constraints but replace the objective by $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) + 4x_2$. At $\mathbf{a}$, verify the KKT first-order conditions, identify an active zero-multiplier constraint, and use SONC to decide whether $\mathbf{a}$ can be a local minimizer of the modified problem.

2. It is given that $\mathbf{x}^* = (4, 0)^T$ is a global maximizer of

$$\begin{aligned} \max \quad & 2x_1 + x_2 \\ \text{s.t.} \quad & h_1(\mathbf{x}) = -x_1 + x_2 \leq 0, \\ & h_2(\mathbf{x}) = (x_1 - 4)^3 + 4x_2 \leq 0, \\ & h_3(\mathbf{x}) = -x_1 \leq 0, \quad h_4(\mathbf{x}) = -x_2 \leq 0. \end{aligned}$$

- (a) Rewrite the objective as a minimization. Do the KKT first-order conditions hold at $\mathbf{x}^*$? Explain why your answer does not contradict KKT FONC.
- (b) Compute $T_{\text{lin}}(\mathbf{x}^*)$ and $T_{\mathcal{S}}(\mathbf{x}^*)$. Verify the tangent-cone first-order necessary condition and identify a direction that is linearly feasible but is not tangent.

3. Consider

$$\begin{aligned} \min \quad & f(\mathbf{x}) = -x_3^2 \\ \text{s.t.} \quad & h_1(\mathbf{x}) = x_1^2 + x_2^2 - x_3 \leq 0, \\ & h_2(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 - 1 \leq 0. \end{aligned}$$

- (a) Prove that a global minimizer exists, that LICQ holds at every feasible point, and find all global minimizers.
- (b) At the minimizer, find the KKT multipliers and the critical cone.
- (c) Compute $\mathbf{H}_L$ and verify SOSC. Explain why the singularity of the ambient $\mathbf{H}_L$ does not prevent this certificate.

4. Consider

$$\begin{aligned} \min \quad & f(\mathbf{x}) = x_1 - x_2 \\ \text{s.t.} \quad & g_1(\mathbf{x}) = x_1^2 + x_2^2 - 2 = 0, \\ & g_2(\mathbf{x}) = x_2 - x_3^3 = 0. \end{aligned}$$

- (a) Find $\alpha > 0$ with $|x_i| \leq \alpha$ for every feasible point and $i = 1, 2, 3$. Explain why a global minimizer exists.
- (b) Prove LICQ at every feasible point and find all global minimizers.
- (c) Find all KKT points and their multipliers. Use critical cones and reduced Hessians to classify the points whenever second-order information gives a conclusion.
- (d) For the remaining points, use the local feasible parametrization

$$\mathbf{x}_{\pm}(t) = (\pm\sqrt{2-t^6}, t^3, t)^T$$

to decide whether they are local minima, local maxima, or saddle points relative to the feasible set.